Reproducibility Budgets for Open-Weight LLM Evaluation: A Generalizability Theory Study of 7–9B Instruction-Tuned Models

Anonymous ACL submission

Abstract

Large Language Model (LLM) evaluation results vary across reproductions, but variance sources interact, and their relative importance shifts with the benchmark format. We apply Generalizability theory (G-theory) in a fully crossed design with up to seven facets across open-weight instruction-tuned models and five benchmarks. Under binary correctness scoring, a recommended 3-prompt $\times$ 3-seed design achieves $G \geq 0.80$ with 178 items—a 3.1$\times$ item reduction over single-condition designs at the cost of 2.9$\times$ more total evaluations—a tradeoff that favors multi-facet designs when item construction is costly. Variance decomposition reveals that models differ primarily in which items they find difficult rather than in overall ability, and that the variance structure shifts qualitatively between multiple-choice and free-form benchmarks. Preliminary validation across model scales and architecture families confirms structural stability. We release an open-source G-theory pipeline and decision-study (D-study) budget calculator for evaluation design optimization[1].

1 Introduction

In our analysis of five benchmarks, item selection drives 34–38% of evaluation variance on multiple-choice tasks while prompt wording is negligible (<3%). On the free-form reasoning task Grade School Math 8K (GSM8K), the structure shifts: model$\times$prompt interaction alone accounts for approximately 9%, while item selection drops to 17%. These numbers, drawn from a factorial analysis of approximately 2.3 million evaluation records, reveal that the variance structure of LLM evaluation is not a fixed property of the pipeline but shifts qualitatively with the response format. Current evaluation practice treats variance sources as independent nuisance factors and controls them one at a time: one study perturbs prompts, another varies precision, a third changes item subsets. This piecemeal approach cannot capture interactions between sources, nor can it predict how the relative importance of each source changes across evaluation settings. Leaderboard rankings built on single-condition evaluations are therefore sensitive to methodological choices that remain unreported and uncontrolled (Rodriguez et al., 2021; Dehghani et al., 2021). No study has simultaneously crossed infrastructure and methodology facets in a unified factorial design across benchmarks to decompose these sources, quantify their interactions, and compare the resulting variance structure.

Existing studies examine individual variance sources in isolation. Sclar et al. (2024) and BrittleBench (Romanou et al., 2026) documented 50–76% accuracy swings from prompt formatting; Yuan et al. (2025) showed up to 9% variation from numerical precision; Pezeshkpour and Hruschka (2024), Dodge et al. (2020), and Henderson et al. (2018) documented ordering and hyperparameter sensitivity—each holding other factors fixed. Messing (2026) applied G-theory to LLM evaluation across five methodology facets on a single benchmark (judge identity: 43.8%), but excluded infrastructure facets and cross-benchmark comparison. Madaan et al. (2024) proposed per-factor metrics without crossing factors. Item Response Theory (IRT) approaches (Maia Polo et al., 2024; Lalor et al., 2019; Zhou et al., 2026) optimize item selection but ignore prompt, seed, and ordering facets. Bouthillier et al. (2021) established analysis-of-variance (ANOVA) based variance decomposition for ML reproducibility. Three gaps remain: no study has (a) crossed infrastructure and methodology facets in a unified factorial design, (b) compared variance structures across response formats, or (c) translated variance components into design-dependent reproducibility budgets.

[1] Code and data are available at `https://github.com/anonymous-nlp-2026-2/nlp_variance_gtheory`.

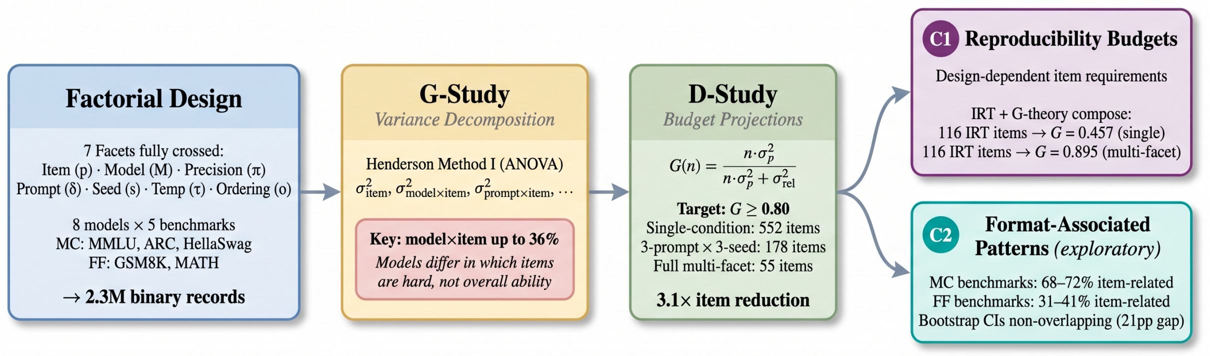

Figure 1: Overview of the G-theory evaluation pipeline. A factorial design crossing seven facets across eight models and five benchmarks produces 2.3M binary records. G-study variance decomposition via Henderson Method I reveals that model×item interaction accounts for up to 36% of variance. D-study projections translate variance components into design-dependent item budgets: a 3-prompt × 3-seed design achieves $G \geq 0.80$ with 178 items (3.1× reduction). Two contributions emerge: reproducibility budgets (C1) and format-associated variance patterns (C2).

Through a factorial G-theory study (Shavelson and Webb, 1991; Brennan, 2001) crossing up to seven facets across eight open-weight 7–9B parameter models and five benchmarks (Massive Multitask Language Understanding [MMLU], AI2 Reasoning Challenge [ARC-Challenge], HellaSwag, GSM8K, MATH), totaling approximately 2.3 million records, we find that evaluation variance has a stable but format-dependent internal structure. D-study projections show that a recommended 3-prompt × 3-seed design achieves $G \geq 0.80$ with 178 items (1,602 evaluations per model), a 3.1× reduction from 552 single-condition items; the full 1,296-condition design further reduces to 55 items as a theoretical ceiling (Figure 2). Model×item interaction (up to 36%) indicates that models differ primarily in *which items* they find difficult (Pezeshkpour and Hruschka, 2024; Maia Polo et al., 2024).

Our contributions:

1. **Design-dependent reproducibility budgets.** A recommended 3-prompt × 3-seed design achieves $G \geq 0.80$ with 178 items (1,602 evaluations per model), a 3.1× reduction over single-condition designs (552 items); the full 1,296-condition design reduces to 55 items (71,280 evaluations). Under binary scoring, BF16 and FP16 require no separate control (<0.1% interaction). IRT item selection and G-theory protocol optimization compose: IRT-selected 116 items yield $G = 0.457$ single-condition but $G = 0.895$ multi-facet (Table 7).

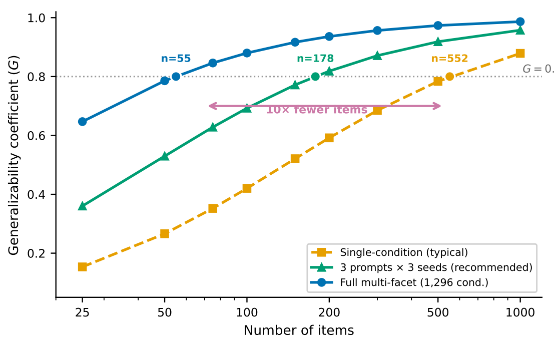

Figure 2: D-study projections for MMLU (single-model baseline). A recommended 3-prompt × 3-seed design reduces the budget from 552 to 178 items for $G \geq 0.80$; the full multi-facet design reduces further to 55 items.

2. **Format-associated variance patterns (exploratory).** Multiple-choice (MC) benchmarks concentrate 68–72% of variance in item-related components; free-form (FF) benchmarks disperse to 31–41% (item-level $d = 1.73$; benchmark-level confirmation limited by sample size). This hypothesis-generating observation warrants confirmation with larger benchmark samples.

We release an open-source G-theory pipeline and D-study budget calculator for evaluation design optimization.

2 Related Work

Variance Sources in LLM Evaluation. Bouthillier et al. (2021) introduced ANOVA-based variance decomposition for ML reproducibility, establishing that design choices account for a

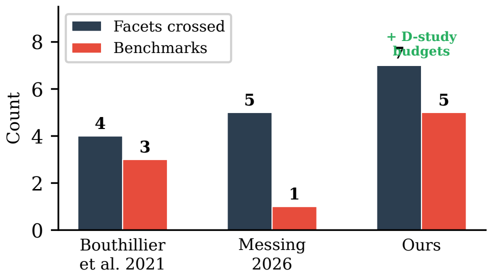

Figure 3: Scope comparison with prior multi-facet variance studies. Our work uniquely combines broad facet crossing, multi-benchmark coverage, and D-study budget projections.

Table 1: Positioning relative to prior multi-facet variance studies. "–" = not addressed.

	Bouthillier et al. (2021)	Messing (2026)	Ours
Facets crossed	4	5	7
Design	Factorial	Factorial	Factorial
Method	ANOVA	G-theory	G-theory
D-study budgets	–	–	✓
Models	ML models	6 LLMs	8 LLMs
Benchmarks	3	1	5
Cross-format	–	–	MC+FF

substantial share of benchmark variance. Subsequent work isolated individual sources: Yuan et al. (2025) documented up to 9% accuracy variation from numerical precision; Sclar et al. (2024) and BrittleBench (Romanou et al., 2026) found 50–76% accuracy swings from prompt formatting; Pezeshkpour and Hruschka (2024) and Bui et al. (2025) documented ordering and seed sensitivity. Messina and Scotta (2026) formalized CUDA nondeterminism from floating-point non-associativity. Each study holds other factors fixed, precluding interaction estimation. Madaan et al. (2024) proposed per-factor metrics without crossing factors; holistic evaluation frameworks (Liang et al., 2023; Gao et al., 2024) reduce implementation variance but do not decompose it. Our work crosses seven facets simultaneously, enabling interaction estimation and cross-format comparison.

IRT and Efficient Benchmarking. IRT models item difficulty and discrimination to optimize benchmark efficiency (Lalor et al., 2019; Rodriguez et al., 2021). Maia Polo et al. (2024) and Zhou et al. (2026) developed IRT-based subsampling for efficient evaluation; Perlitz et al. (2023) and Kipnis et al. (2025) explored related compression strategies. These approaches optimize item selection but do not model variance from prompt, seed, or ordering facets. IRT-selected items under a single methodological condition can therefore underestimate required item counts—our results show IRT's 116 items yield $G = 0.457$ single-condition but $G = 0.895$ multi-facet, demonstrating that item selection and protocol optimization are orthogonal.

G-Theory and Positioning. G-theory (Shavelson and Webb, 1991; Brennan, 2001) provides the canonical framework for multi-facet reliability analysis, with extensions to binary data (Mushquash and O'Connor, 2006; Jiang and Skorupski, 2018). We operationalize differential item functioning (Holland and Wainer, 1993; Clauser and Mazor, 1998) as model×item interaction. Recent psychometric perspectives on LLM evaluation address construct validity (Kearns, 2026) and measurement properties (Ye et al., 2025). Closest to our work, Messing (2026) applied G-theory to LLM evaluation, decomposing five methodology facets on a single benchmark (judge identity: 43.8%). We extend this in three directions: (i) replacing judge identity with infrastructure facets and additional methodology facets; (ii) crossing facets across five benchmarks for cross-format comparison; and (iii) translating variance components into reproducibility budgets via D-study projections. Table 1 summarizes these distinctions.

3 Method and Experimental Design

3.1 Generalizability Theory

Generalizability theory (G-theory) treats each measurement as one observation from a universe of admissible conditions defined by the crossed facets of an evaluation design (Shavelson and Webb, 1991; Brennan, 2001). The *object of measurement* is the entity whose consistency is of interest. In our setting, test items (p) serve as the object of measurement, and the goal is to determine how reliably item-level scores generalize across the methodological conditions under which they are collected.

A *G-study* decomposes total observed variance into components σ^2_x for each facet x and their interactions. Each component quantifies how much variability that source contributes to the overall measurement. Large interaction components (e.g., σ^2_{pM} for item-by-model) indicate that facets do not affect items uniformly but instead create differential difficulty profiles. We estimate variance components via Henderson Method I, an unbiased ANOVA-based estimator for balanced designs (Brennan, 2001, Ch. 3), with 1,000-resample bootstrap confidence intervals (Efron and Tibshirani, 1993) (item-level resampling with replacement, preserving the fully-crossed structure across all other facets) (Li, 2023).

A *D-study* projects the generalizability coefficient under hypothetical designs with different numbers of conditions per facet:

$$G_{\text{item}} = \frac{\sigma_p^2}{\sigma_p^2 + \sigma_{\text{rel}}^2} \tag{1}$$

where $\sigma_{\text{rel}}^2 = \sum_i \sigma_{p\times f_i}^2 / n_{f_i} + \sigma_{\text{residual}}^2 / \prod_i n_{f_i}$ aggregates all item-by-facet interaction variances, each divided by the number of conditions for the corresponding facets (Brennan, 2001, Ch. 4). A D-study projects reliability for a test averaging over n_p items via

$$G(n_p) = \frac{n_p\,\sigma_p^2}{n_p\,\sigma_p^2 + \sigma_{\text{rel}}^2} \tag{2}$$

enabling independent verification of all budget projections in Table 7. This prospective capability distinguishes G-theory from classical ANOVA: rather than only attributing variance post hoc, a D-study enables optimization of evaluation designs *before* data collection (Shavelson et al., 1989). Figure 4 illustrates the practical impact: multi-facet designs achieve $G \geq 0.80$ with 55 items versus 552 under single-condition protocols. We define *item-related variance* as the sum of the item main effect (σ_p^2) and the model×item interaction (σ_{pM}^2). The latter captures differential item functioning (DIF): items whose difficulty ranking changes across models (Brennan, 2001, Ch. 4). Because DIF is operationally an item-level property—it reflects model-specific item difficulty profiles rather than a global model ability shift—aggregating $\sigma_p^2 + \sigma_{pM}^2$ provides a unified measure of item-driven variance. We report unbundled components throughout to ensure transparency.[2]

3.2 Experimental Design: MMLU

We design a fully crossed factorial experiment (exp-001) that simultaneously varies seven facets of LLM evaluation on MMLU (Figure 5). Table 2 summarizes the complete design matrix.

[2] Notation: p = item, M = model, π = precision, δ = prompt, s = seed, τ = temperature, o = ordering, σ_x^2 = variance component for source x.

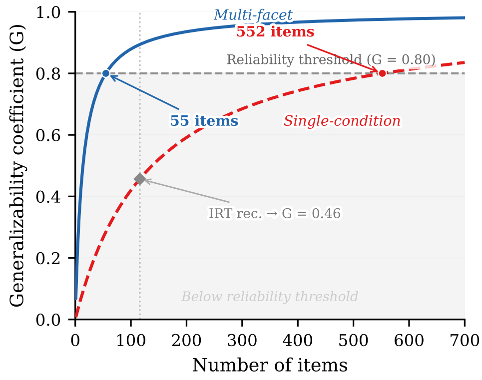

Figure 4: D-study projection: multi-facet designs (blue) reach the $G \geq 0.80$ reliability threshold at 55 items, while single-condition protocols (red) require 552—a 10× reduction in test length.

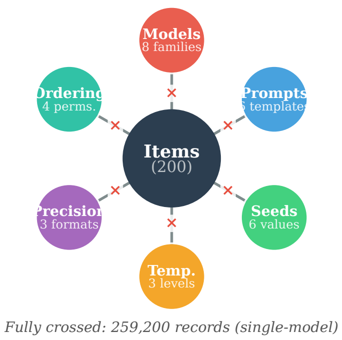

Figure 5: Fully crossed factorial design. Each of six facets is crossed with all 200 items, yielding 259,200 observations per model.

Facets. Items (p): 200 MMLU questions via stratified sampling across 10 subjects (20 per subject, four domains: STEM, humanities, social sciences, professional; coefficient of variation [CV] = 3.1%). Models (M): eight open-weight 7–9B instruction-tuned models (Table 3), treated as a quasi-random facet (§3.6). Numerical precision (π): BFloat16 (BF16), Float16 (FP16), Float32 (FP32). Prompt template (δ): six surface perturbations (Sclar et al., 2024) varying layout and wording (Appendix G). Random seed (s): six fixed values. Sampling temperature (τ): 0.0, 0.3, 0.7. Answer ordering (o): four permutations of few-shot demonstrations (Pezeshkpour and Hruschka, 2024).

Sample sizes. The cross-model BF16 balanced subset yields 460,800 records (8 models × 200 items × 6 prompts × 6 seeds × 2 temperatures × 4 orderings). A single-model analysis (Llama) spans all three precisions, yielding 259,200 binary ob-

Table 2: Fully crossed G-theory design. Exp-001 evaluates MMLU across all facet levels; exp-002 extends to five benchmarks with reduced crossing. †BF16 balanced subset (2 temperatures: 0.0, 0.7) for cross-model analysis; the full single-model design uses all 3 levels.

Facet	Levels	exp-001	exp-002
Item (p)	Stratified sample	200	200×5
Model (M)	8 architecture families	8	8
Precision (π)	BF16 / FP16 / FP32	3	1
Prompt (δ)	Paraphrases	6	6
Seed (s)	Fixed set	6	6
Temperature (τ)	0.0 / 0.3 / 0.7	3	2
Ordering (o)	Demo permutations	4	4
Total records		460,800†	2,304,000

servations across 1,296 fully balanced conditions. The BF16 subset serves as the primary cross-model sample because BF16 is the standard inference precision for these architectures.

3.3 Cross-Benchmark Extension

Exp-002 extends to five benchmarks: MMLU, ARC-Challenge, HellaSwag (MC), GSM8K, and MATH (FF). The design crosses 8 models × BF16 × 2 temperatures (0.0, 0.7) × 6 prompts × 6 seeds × 4 orderings × 200 items, totaling 2,304,000 records. This reduced crossing fixes precision at BF16 to concentrate power on format differences. Ordering operates only on MMLU ($\sigma^2_{\text{ordering}} = 0$ for other benchmarks, retained for design balance). Subset analysis confirms this preserves the full design's variance structure ($\Delta G = 0.0008$; Appendix A).

3.4 Evaluation Protocol

All evaluations use generation-based scoring (free-text response with answer extraction) rather than log-likelihood ranking, following Hua et al. (2025). Binary correctness (1/0) is the primary metric; text-exact-match is secondary. Inference uses vLLM 0.9.2 without batch-invariant kernels, preserving CUDA-level nondeterminism as a measurable variance source (Messina and Scotta, 2026; He and Thinking Machines Lab, 2025). Few-shot protocols: 5-shot MMLU/ARC, 4-shot MATH, 8-shot GSM8K, 10-shot HellaSwag.

3.5 Binary Data Assumptions

Standard G-theory assumes continuous data (Shavelson and Webb, 1991); our binary scores violate this. Henderson Method I produces unbiased estimates for balanced designs regardless of distribution (Brennan, 2001, Ch. 12), and restricted

Table 3: Models used in all experiments. RMS = RMSNorm, LN = LayerNorm.

Short name	Family	Params	Norm
Llama-3.1-8B	Llama-3	8B	RMS
Gemma-2-9B	Gemma-2	9B	RMS
Mistral-7B-v0.3	Mistral	7B	RMS
Qwen3-8B	Qwen-3	8B	RMS
Yi-1.5-9B	Yi-1.5	9B	RMS
OLMo-2-7B	OLMo-2	7B	RMS
InternLM2.5-7B	InternLM-2	7B	RMS
DeepSeek-7B	DeepSeek	7B	RMS

maximum likelihood (REML) differs by <0.5pp. A probit simulation, generalized linear mixed model (GLMM) complement ($\rho = 1.0$), and accuracy-range analysis confirm robustness (Appendix A).

All findings are conditioned on binary correctness scoring; the text-exact-match comparison in §4.1 ($G = 0.007$ vs. 0.936) illustrates how scoring can reshape variance structure. Partial-credit analysis on GSM8K confirms G_{item} drops from 0.86 to 0.77 (mean $\Delta G = -0.135$; Appendix A), suggesting limited transferability to finer-grained rubrics.

3.6 Model Facet Specification

Our target universe comprises open-weight, instruction-tuned models in the 7–9B parameter range (Table 3); results may not generalize to API-based, base, or >9B models. The eight models are treated as a quasi-random facet (Meredith, 1993; Putnick and Bornstein, 2016). We report G under random-model assumptions, the more conservative specification: treating models as fixed yields G_{fixed} 0.951–0.990 vs. G_{random} 0.863–0.896 (increase of 0.077–0.113; Table 10), because model×item interaction (30–36%) enters relative error variance. Consequently, D-study budgets are upper bounds. LOMO analysis preserves component rankings across all five benchmarks (Appendix A); four-model subset checks match within $|\Delta G| = 0.035$ (§4.5).

4 Experiments

All experiments use open-weight 7–9B parameter models evaluated with vLLM 0.9.2 on a single NVIDIA RTX PRO 6000 GPU (96GB) with binary correctness scoring.

4.1 Single-Model Baseline

We first establish the variance structure within a single model (Llama-3.1-8B-Instruct) on MMLU to isolate methodological facet contributions before

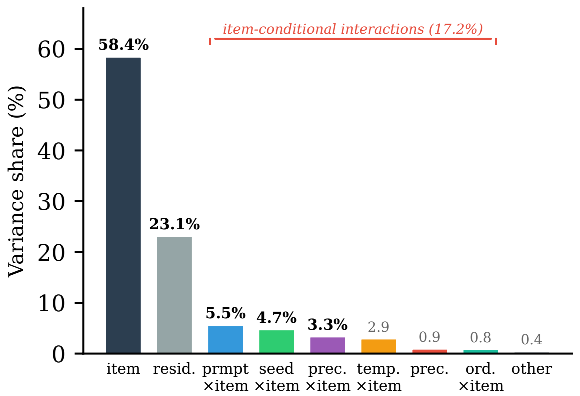

Figure 6: Single-model variance decomposition (Llama-3.1-8B, MMLU). Item difficulty dominates (58.4%); all main facet effects are <1%. Item-conditional interactions total 17.2%.

introducing cross-model variation.

Design. The G-study crosses six facets: 200 items × 3 numerical precisions (BF16, FP16, FP32) × 6 prompt templates × 6 random seeds × 3 temperature levels (0.0, 0.3, 0.7) × 4 answer orderings, yielding 259,200 binary observations. Variance components are estimated via Henderson Method I with bootstrap confidence intervals (1,000 resamples).

Results. Figure 6 shows the variance decomposition. Item difficulty dominates at 58.4%, with residual at 23.1%. All main facet effects are negligible (<1% each). The interesting structure is in item-conditional interactions (17.2% total): prompt×item (5.5%) is largest, indicating items are differentially sensitive to specific facets. The generalizability coefficient $G_{\text{item}} = 0.936$ shows 200 items yield highly reliable rankings; D-study projection gives $G \geq 0.80$ with 55 items.

For text-exact-match scoring (requiring verbatim string matches), the variance structure shifts dramatically: temperature dominates at 79.7% of variance, and G_{item} collapses to 0.007. This metric-conditional divergence motivates our exclusive use of binary correctness for the remaining analyses.

4.2 Cross-Model Variance

Extending to eight models (BF16 subset, 460,800 records), Table 5 shows that item difficulty drops from 58.4% to 37.9%, while model×item interaction emerges at 30.2% (up to 36% across benchmarks, §4.3). The model main effect is 3–5% (MC)

Table 4: LOMO stability across five benchmarks (8 models × 5 benchmarks = 40 subsets).

Benchmark	G_{full}	CV (%)	Max $\lvert\Delta G\rvert$	Most impactful
MMLU	0.896	0.49	0.019	InternLM2.5-7B
ARC	0.892	0.94	0.026	Llama-3.1-8B
HellaSwag	0.872	1.09	0.028	Llama-3.1-8B
GSM8K	0.863	1.25	0.030	Gemma-2-9B
MATH	0.888	1.20	0.029	InternLM2.5-7B

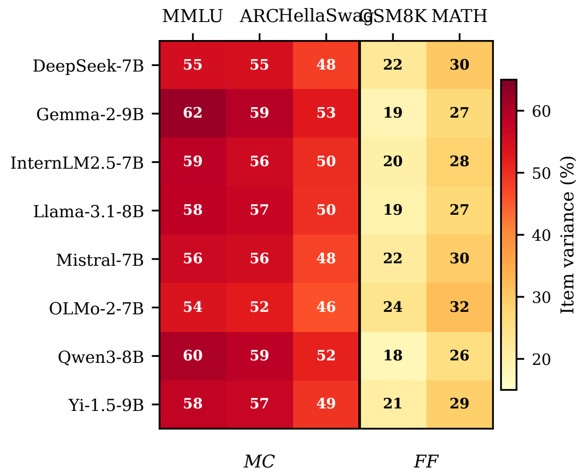

Figure 7: Item variance share (%) by model and benchmark. MC benchmarks (left) show higher item variance (46–62%) than FF benchmarks (right, 18–32%) across all eight models.

and 9–11% (FF)[3], confirming that models differ primarily in *which items* they find difficult rather than in overall ability level (contamination implications in §6). Leave-one-model-out (LOMO) analysis (Table 4) confirms this structure is stable: no single model removal changes G by more than 0.030 across all five benchmarks.

4.3 Exploratory: Format-Associated Variance Patterns

We extend to five benchmarks (three MC, two FF) using exp-002. Table 6 reveals a format-associated pattern: MC benchmarks concentrate item-related variance (item + model×item DIF) at 68–72%, while FF benchmarks disperse to 31–41% ($n = 5$; permutation $p = 0.10$; item-level $d = 1.73$, benchmark-level CI spans zero); we report this as a descriptive observation. This does not extend to item budgets: HellaSwag (MC, 118 items) exceeds ARC (98) and MATH (101), showing format alone does not predict cost (Figure 9). The design is not fully symmetric: ordering applies only to MMLU, potentially confounding format

[3] Ranges rounded to nearest integer; exact values are 2.8–5.4% (MC) and 8.5–10.7% (FF) per Table 6.

Table 5: Variance decomposition for the cross-model MMLU design (8 models, BF16, 460,800 records). Components estimated via Henderson Method I. $G_{\text{item}} = 0.896$.

Component	Var.%	95% CI
item_id	**37.9**	[33.0, 42.4]
model×item	30.2	[26.9, 33.3]
residual	25.0	[22.9, 27.0]
model	2.8	[1.7, 4.3]
model×prompt	1.6	[1.2, 2.1]
prompt×item	0.9	[0.8, 1.1]
temp.×item	0.6	[0.4, 0.8]
seed×item	0.5	[0.4, 0.7]
ordering×item	0.3	[0.2, 0.4]
all others (each)	<0.1	—

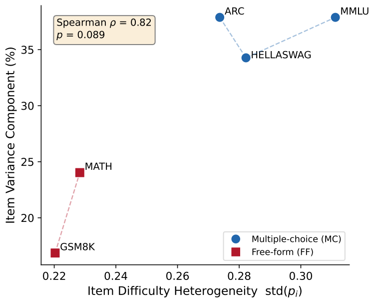

Figure 8: Item difficulty heterogeneity (std(p_i)) versus `item_id` variance share across five benchmarks. MC benchmarks (filled) exhibit higher difficulty dispersion and item variance than FF benchmarks (open). Spearman $\rho = 0.82$.

Table 6: Format-associated variance structure across five benchmarks (exp-002, 8 models). Variance components (%) per benchmark; MC = multiple-choice, FF = free-form. Bold = dominant non-residual component. †For MC, reports prompt×item interaction (prompt main effect <1%); for FF, reports prompt main effect (item-conditional interactions absorbed into residual).

	MC			FF	
	MMLU	ARC	Hella.	GSM	MATH
item	**37.9**	**37.9**	34.3	16.9	24.0
mod.×item	30.2	33.7	**36.0**	14.6	17.3
model	2.8	3.1	5.4	10.7	8.5
prompt†	0.9	1.4	2.1	5.4	0.8
mod.×prmpt	1.6	8.0	0.2	8.5	1.7
residual	25.0	15.4	21.4	39.9	43.9
all others	1.6	0.5	0.6	4.0	3.8

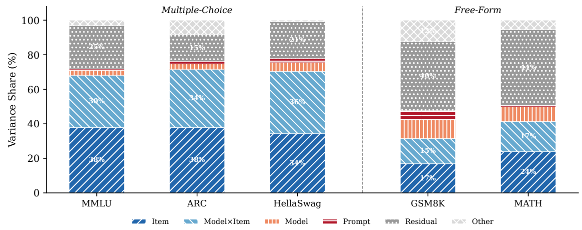

Figure 9: Variance decomposition (Henderson Method I) across five benchmarks and eight models. Multiple-choice benchmarks share an item-dominated structure; HellaSwag is the exception where model×item (36.0%) exceeds item difficulty (34.3%). Free-form benchmarks (GSM8K, MATH) distribute variance across item, model, and interaction components with no single dominant source.

comparisons. Figure 8 shows item difficulty heterogeneity (std(p_i)) predicts item variance share ($\rho = 0.82$; multi-scale support in Appendix D). Per-benchmark details, including residual variance regression (Appendix F) and subject-level decomposition within MMLU (Appendix E), are in Appendix C. All findings are conditional on binary scoring (§3.5).

4.4 Precision Separability

Under binary scoring, BF16 and FP16 are functionally interchangeable (<0.1% interaction (Yuan et al., 2025)); under text-exact-match, a precision×temperature interaction emerges at 1.3%, consistent with §4.1. Precision need not be controlled for binary evaluation at BF16/FP16 (Shanmugavelu et al., 2024).

4.5 Design-Dependent Reproducibility Budgets

How many items does a given evaluation design need to achieve a target reliability? We translate variance components into item budgets via D-study projections.

Table 7 compares evaluation designs at the $G \geq 0.80$ threshold using single-model (Llama-3.1-8B-Instruct) MMLU variance components. The multi-facet design (1,296 conditions per item) achieves $G = 0.936$ at 200 items and requires only 55 items for $G \geq 0.80$. A single-condition design reaches only $G = 0.592$ at 200 items and requires 552 items—a 10.0× budget ratio stable across subset sizes (CV = 1.9%).

The cost–reliability tradeoff across intermediate designs (Table 8) reveals that the steepest gain occurs at the minimal upgrade level (2 prompts): G_{200} jumps from 0.592 to 0.692 at 2× cost. A minimal sufficient design for $G \geq 0.80$ is 3 prompts × 3 seeds (9 conditions, $n_{G \geq 0.80} = 178$). Beyond 32 conditions, adding orderings provides negligible benefit (ordering×item <1%), confirming that prompt and seed diversity are the primary reliability drivers.

Table 7: D-study reproducibility budgets under different evaluation designs (single-model, MMLU). Total evals = items × conditions.

Design	Cond.	$n_{G\geq.80}$	Total	G_{200}	G_{116}
Multi-facet	1,296	**55**	71,280	**.936**	—
Single-cond.	1	552	552	.592	.457
IRT+multi	1,296	—	150,336	—	.895‡
IRT baseline	—	116	116	—	.457†

† PSN-IRT 116 items (Zhou et al., 2026) under single-condition: $G = 0.457$. ‡ Same 116 items under multi-facet: $G = 0.895$.

Table 8: Intermediate D-study designs ordered by increasing conditions per item. G_{200} and $n_{G\geq0.80}$ are D-study projections from exp-001 variance components.

Design	Conditions	G_{200}	$n_{G\geq.80}$
Single-condition	1	.592	552
Minimal upgrade (2p)	2	.692	357
Easy upgrade (2p×2s)	4	.766	245
Moderate (3p×3s)	9	.818	178
Standard (4p×4s×2t)	32	.869	121
Enhanced (+4 orderings)	72	.871	119
Full multi-facet	1,296	.936	55

Zhou et al. (2026)'s IRT-selected 116 items yield $G = 0.457$ under single-condition but $G = 0.895$ under multi-facet conditions, demonstrating that IRT item selection and G-theory protocol optimization are orthogonal and complementary. An internal consistency check using four-model subsets confirms projection stability ($|\Delta G| \leq 0.029$ for MC, ≤ 0.065 for FF; Table 11); the corresponding ∼23% budget increase is consistent across all five benchmarks (Figure 10).

The design choice depends on the evaluation bottleneck: the full 1,296-condition design minimizes items (55); the recommended 3×3 balances cost and reliability (178 items, 1,602 evaluations); single-condition requires 552. The efficiency ratio is format-associated: MC benchmarks show 11–13× (mean 12.0×), while FF benchmarks show 21–22× (mean 21.5×). Cross-model projections (Table 9) show budget ratios all exceeding 10× (bootstrap CIs), with multi-facet budgets of 93–128 items.

Projection assumptions. Cross-model budgets increase to 93 items because model×item interaction enters relative error variance. G_{item} is stable across benchmarks (0.863–0.896, CV = 1.4%; Appendix B). All projections assume the 2024–2025 model generation; §4.6 provides preliminary support.

Table 9: Cross-model D-study reproducibility budgets with 95% bootstrap CIs. $n_{G\geq0.80}$: items needed. Multi = multi-facet (2,304 cond.); Total = items × 2,304. Single = single-condition (total evals = item count).

	Multi-facet (2,304 cond.)			Single-cond.		
	Items	95% CI	Total	Items	95% CI	Ratio [95% CI]
Multiple-choice						
MMLU	93	[76, 116]	214K	1,216	[1004, 1491]	13.1 [12.5, 13.7]
ARC	98	[71, 139]	226K	1,078	[793, 1512]	11.0 [10.7, 11.4]
Hella.	118	[93, 151]	272K	1,403	[1120, 1790]	11.9 [11.5, 12.3]
Free-form						
GSM8K	128	[103, 162]	295K	2,768	[2230, 3531]	21.6 [20.7, 23.0]
MATH	101	[86, 122]	233K	2,160	[1853, 2631]	21.4 [20.2, 22.5]

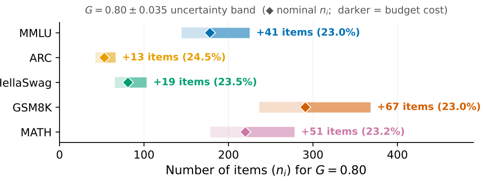

Figure 10: Effect of $|\Delta G| = 0.035$ estimation error on item budgets. The ∼23% increase is consistent across benchmarks.

4.6 Scale Validation

To test whether the variance structure generalizes beyond 7–9B models, we conduct within-family and cross-family analyses. Qwen2.5-72B-Instruct on MMLU shows item difficulty at 83.2% ($G_{\text{item}} = 0.979$), with prompt×item (6.8%) as the dominant interaction—the same ordering observed at 7B. A three-model crossed design (Qwen2.5 3B/7B/72B) preserves the model×item dominant interaction (38.5%) and item main effect (35.7%). Cross-family, Llama-3.1-70B-Instruct yields item difficulty at 74.5% ($G_{\text{item}} = 0.957$), confirming that item difficulty dominates across a 24× parameter range and two architecture families.

Robustness checks. Three MMLU resamples yield CV = 3.1%; treating models as fixed increases G by 0.077–0.113, confirming conservative budgets (§3.6; Appendix A).

5 Conclusion

A 3-prompt × 3-seed design reduces item requirements from 552 to 178 for $G \geq 0.80$, confirmed across five benchmarks under binary scoring. The variance structure is format-dependent: MC benchmarks concentrate 68–72% in item-related components while FF benchmarks disperse to 31–41%, suggesting per-format budget calibration. We recommend multi-facet validation (at minimum 3×3) as standard practice; leaderboard operators should average over prompt and seed conditions.

6 Limitations

Our design covers eight 7–9B instruction-tuned models; preliminary 72B validation confirms consistent variance structure, but broader scale and temporal coverage is needed. The benchmark sample ($n = 5$) limits statistical power for format-associated observations ($p = 0.10$). All D-study budgets assume binary scoring; partial-credit analysis on GSM8K shows G drops from 0.86 to 0.77, suggesting limited transferability to finer-grained rubrics.

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A Additional Robustness Analyses

We report nine robustness analyses that verify the stability of the variance decomposition under alternative analytical choices.

GLMM complement. A generalized linear mixed model (GLMM) with marginal logit link produces variance component rankings that perfectly preserve all seven grouped component orderings ($\rho = 1.0$) relative to Henderson Method I (Jiang and Skorupski, 2018; Tuerlinckx et al., 2006). This confirms that the linear variance decomposition is robust to the binary outcome distribution.

Accuracy-range stability. The `item_id` variance share remains stable across subsets with mean accuracy from 0.62 to 0.72 ($r = -0.048$). This rules out distortion from binary floor or ceiling effects at observed accuracy levels.

Estimator robustness. We compare Henderson Method I with restricted maximum likelihood (REML-EM, 500 iterations, practical convergence $\Delta < 0.003$ percentage points). All variance components differ by <0.5 percentage points between estimators. The largest discrepancy occurs on the residual component (0.47pp); all substantive components (item, model$\times$item, model) differ by <0.3pp. Henderson Method I is preferred for our balanced designs because it produces unbiased estimates without distributional assumptions (Brennan, 2001, Ch. 12).

Probit simulation. A probit simulation matched to our 1,296-condition design confirms that Henderson Method I recovers variance component rankings from dichotomized data (Spearman $\rho = 0.732$ between binary and continuous rankings, excluding residual; `item_id` dominant in all 100 replications; top-5 ordering correct in 79% of replications). Bias analysis by difficulty stratum shows $|\text{bias}| = 0.28$–0.37 for items with $p \in [0.5, 0.8]$, where the Bernoulli variance ceiling $p(1-p)$ is largest. This analysis covers the observed accuracy range ($p \in [0.3, 0.9]$); extreme-difficulty items ($p < 0.3$ or $p > 0.9$) are rare in our sample (<8% of items) and may exhibit larger deviations. Henderson Method I absorbs this Bernoulli-inherent variance into the residual, explaining inflated bias near medium difficulty; systematic component ordering is robustly recovered across all replications. Binary scoring inflates residual variance by absorbing the Bernoulli $p(1-p)$ ceiling, which enters σ^2_{rel} and increases D-study item requirements. Within a matched probit simulation, binarization increases D-study budgets by approximately 36% (99 $\rightarrow$ 135 items for $G \geq 0.80$), driven by residual inflation (16% $\rightarrow$ 41%) and item variance compression (50% $\rightarrow$ 32%). Facet$\times$item interactions—the components entering σ^2_{rel}—are best preserved under binarization (MAPE = 12.8%, range 8.7–17.5%), while residual variance inflates as expected from Bernoulli $p(1-p)$ noise (MAPE = 156%). D-study budgets derived from binary data are therefore conservative: continuous-scored designs would require equal or fewer items for the same G threshold. Null calibration (ratio $= 1.0008$) confirms that this conservatism does not distort relative component magnitudes.

CI method comparison. Bootstrap (1,000 item-level resamples with replacement) and delete-1 jackknife confidence intervals agree closely: width ratios range from 0.96 to 1.04 across all variance components. Bootstrap CIs capture item-sampling variability conditional on these 8 specific models; model-level stability is assessed separately via LOMO (below). The agreement indicates that the dependence structure introduced by item-level bootstrap resampling (which creates $\sim$127 unique items per 200-item resample) does not materially distort interval estimates. Stratified resampling by subject (10 subjects $\times$ 20 items) produces CIs within 4% of simple item-level bootstrap, confirming that the within-subject dependence structure does not materially affect interval estimates.

Facet specification sensitivity. Treating temperature and ordering as fixed (rather than random) facets changes G_{item} by only 0.37 percentage points (all-random: 0.820 vs. fixed temperature+ordering: 0.824). The small difference reflects the negligible temperature$\times$item (0.65%) and ordering$\times$item (0.54%) variance shares. The random-facet assumption is therefore robust to facet classification.

Leave-one-model-out (LOMO). Across all 40 leave-one-model-out subsets (8 models $\times$ 5 benchmarks), G_{item} ranges from 0.833 to 0.919, with CV $\leq 1.25\%$ per benchmark. No single model removal changes G by more than 0.030 (GSM8K, removing Gemma-2-9B). Item-related variance rankings are preserved in all 40 subsets. Table 4 (main text) reports per-benchmark stability. This supports treating models as exchangeable within the target universe of 7–9B instruction-tuned models (§3.6).

Inter-model agreement. We computed pairwise Cohen's κ and Fleiss' κ across all model pairs on each benchmark. Mean pairwise κ ranges from 0.326 (MATH) to 0.461 (MMLU); Fleiss' κ ranges from 0.309 (GSM8K) to 0.456 (MMLU). HellaSwag uses six models ($\kappa = 0.332$) due to missing OLMo and Yi data. The fair-to-moderate agreement independently confirms that the model$\times$item interaction (30–36% of variance) reflects genuine differential item functioning: models disagree substantially on which items are difficult, consistent with prior DIF findings in educational measurement (Holland and Wainer, 1993; Clauser and Mazor, 1998).

Scoring metric sensitivity. To probe metric conditionality, we compared binary correctness with step-level partial credit on GSM8K using the full 8-model crossed design (200 items, 2,304 conditions per item). Under partial credit, G_{item} drops from 0.863 to 0.768, with residual variance rising from 39.9% to 64.9% and item variance falling from 16.9% to 10.3%. All eight models individually show $G_{\text{partial}} < G_{\text{binary}}$ (mean $\Delta G = -0.135$, range -0.025 to -0.309), with ΔG magnitude moderated by model accuracy: higher-accuracy models have fewer incorrect responses whose scores shift under partial credit (e.g., OLMo at 70.1% accuracy shows $\Delta G = -0.025$, while Llama at 49.0% shows $\Delta G = -0.309$). The mechanism is clear: 48.1% of incorrect responses have numerically correct arithmetic steps but incorrect problem setup, receiving partial score $= 1.0$ despite binary score $= 0$—this compresses item discrimination and inflates residual variance. Binary scoring amplifies prompt sensitivity by collapsing reasoning quality into pass/fail, confirming that all findings in this paper are metric-conditional.

Facet specification sensitivity: random vs. fixed model. Table 10 reports per-benchmark generalizability coefficients under random and fixed model specifications. The consistent ΔG of 0.077–0.113 confirms that random-model budgets are strictly more conservative across all benchmarks.

Table 10: Generalizability coefficients under random vs. fixed model facet specification across five benchmarks.

Benchmark	G_{random}	G_{fixed}	ΔG
MMLU	0.896	0.984	+0.088
ARC	0.892	0.990	+0.098
HellaSwag	0.872	0.984	+0.113
GSM8K	0.863	0.951	+0.089
MATH	0.888	0.965	+0.077
Mean	0.882	0.975	+0.093

Null calibration. We generate 100 Bernoulli-distributed datasets with item difficulties matched to the empirical distribution but no systematic facet effects. The observed-to-null variance ratio is 1.0008, confirming that Henderson Method I does not introduce systematic bias when applied to binary outcomes. The null simulation also verifies that the item variance share (58.4%) is not an artifact of the Bernoulli $p(1-p)$ ceiling: the null produces item variance of 0.08%, three orders of magnitude below the observed value. Across all 100 replications, the same three components (item, model$\times$item, residual) emerge as the top three (set consistency 100%), with Kendall's $\tau = 0.81$ between binary and continuous rankings.

Reduced crossing validation. To verify that the reduced crossing in exp-002 (BF16 only $\times$ 2 temperatures) does not introduce confounding, we extracted the corresponding subset from the fully crossed exp-001 design. The generalizability coefficient differs by only 0.0008 ($G = 0.9361$ vs. 0.9369), and all non-precision variance components differ by less than 2 percentage points. The 10pp increase in item variance share (58.4% $\rightarrow$ 68.5%) is fully explained by the removal of precision-related components ($\sim$4.5% total), which redistributes across remaining sources. This confirms that the exp-002 design preserves the variance structure of the full crossing.

B Projection Validation Details

Internal consistency check. Table 11 reports consistency checks from four exp-001 models (DeepSeek, Llama, Mistral, Qwen3) to the full eight-model design across all five benchmarks. MC benchmarks show $|\Delta G| \leq 0.029$ (<3.5% error), while FF benchmarks show larger deviations ($|\Delta G| \leq 0.065$, $\sim$7%) due to limited model diversity in the four-model subset. FF predictions consistently underestimate actual G (conservative); MC predictions show mixed directions with small deviations ($|\Delta G| \leq 0.029$).

Table 11: Internal consistency: four-model subset vs. eight-model G_{item}. FF subsets underestimate actual G (conservative); MC shows mixed directions with small deviations.

Benchmark	Predicted G	Actual G	$\lvert\Delta G\rvert$	Error%
Multiple-choice				
MMLU	0.888	0.896	0.009	0.96
ARC	0.904	0.892	0.012	1.38
HellaSwag	0.901	0.872	0.029	3.33
Free-form				
GSM8K	0.803	0.863	0.059	6.88
MATH	0.823	0.888	0.065	7.30
Mean	—	—	0.035	3.97

Budget sensitivity to estimation error. The effect of $|\Delta G| = 0.035$ estimation error on item budgets is shown in Figure 10 (main text). The percentage increase is remarkably stable (23.0–24.5%, mean 23.2%), because the relative budget adjustment $\Delta n/n$ depends on the G-formula ratio rather than on absolute variance components. Absolute Δn ranges from 13 (ARC) to 67 (GSM8K), scaling with the baseline budget.

Cross-benchmark G stability. G_{item} values cluster in a narrow band across all five benchmarks (0.863–0.896, CV = 1.4%), despite qualitatively different variance structures: MC benchmarks concentrate 68–72% in item-related components while GSM8K disperses variance across item (17%), model (11%), and interaction (15%) sources (Table 6). This stability implies that D-study budget recommendations calibrated on one benchmark provide reasonable order-of-magnitude guidance for others. However, the 0.033 range in G_{item} translates to a 71–128 item range for multi-facet $G \geq 0.80$ (Table 9), and the variance compositions differ substantially—benchmark-specific G-studies remain necessary to identify dominant variance sources and optimize condition selection.

C Format-Associated Variance: Per-Benchmark Details

This section supplements the compressed format-associated analysis in §4.3 with per-benchmark details.

MC benchmark exceptions. HellaSwag is the sole MC case where model×item slightly exceeds item (36.0% vs. 34.3%), though their sum (70.3%) remains in the MC range (68–72%). Among MC benchmarks, ARC-Challenge shows an anomalously high model×prompt interaction (8.0% vs. $<$2% for MMLU and HellaSwag), suggesting that science-domain items are differentially sensitive to option formatting across model families.

FF benchmark patterns. Free-form benchmarks differ markedly from MC. On GSM8K, variance distributes across item (16.9%), model×item (14.6%), model (10.7%), and model×prompt (8.5%); prompt main effect is 5.4%. MATH shows an intermediate pattern: item leads at 24.0% with model×item at 17.3%, but residual variance is substantially larger (43.9%). Item-level regression reveals that FF residual variance is predominantly explained by the Bernoulli variance ceiling $p(1-p)$ ($R^2 = 0.47$–0.86; Appendix F): medium-difficulty items maximize binary outcome variance regardless of facet structure, so the elevated FF residuals reflect measurement resolution rather than unmodeled substantive sources.

Difficulty dispersion as variance predictor. Difficulty dispersion (std(p_i)) strongly predicts item variance share ($\rho = 0.82$, $n = 5$ benchmarks; Appendix D; reasoning-heavy subjects show G_{item} as low as 0.614, Appendix E). Format constrains the noise floor, but difficulty heterogeneity determines variance magnitude within it.

Item-level bootstrap stability. Bootstrap resampling (1,000 item resamples per benchmark) confirms that the MC–FF separation is robust to item selection: 95% CIs for item-related variance (item + model×item) are [65.2, 75.0]% (MC) and [28.0, 44.2]% (FF), with no overlap (minimum gap 21.0 percentage points).

D Cross-Benchmark Difficulty Heterogeneity

Item difficulty heterogeneity is shown in Figure 8 (main text). MC benchmarks cluster at higher dispersion (0.27–0.31) and higher item variance (34–38%), while FF benchmarks occupy the lower-left region (0.22–0.23, 17–24%). The cross-benchmark Spearman $\rho = 0.82$ ($p = 0.089$, $n = 5$) is consistent in direction with the within-MMLU correlation ($\rho = 0.806$, $p = 0.005$, $n = 10$ subjects); these are independent analyses at different granularities (benchmarks vs. subjects within MMLU), providing multi-scale support for the item difficulty heterogeneity mechanism discussed in §4.3.

E MMLU Subject-Level Variance Decomposition

Table 12 reports per-subject variance decomposition for 10 MMLU subjects (20 items each, 8 models, 2,304 conditions per item). G_{item} ranges from 0.614 (`high_school_mathematics`) to 0.930 (`international_law`), with mean 0.843. Knowledge-intensive subjects (`international_law`, `clinical_knowledge`, `world_religions`) show `item_id` $>$ 40%, while reasoning-heavy subjects (`high_school_mathematics`, `abstract_algebra`) show model×item $>$ `item_id`, consistent with models differing more in reasoning strategies than in factual recall. The `high_school_mathematics` `item_id` of 10.4% referenced in §4.3 is the lowest across all subjects, below even GSM8K (16.9%).

Table 12: Subject-level variance decomposition within MMLU (20 items per subject, 8 models, 2,304 conditions per item). Sorted by G_{item} descending.

Subject	Acc.	item%	mod.×it.%	G_{item}
international_law	.717	44.4	24.1	.930
clinical_knowledge	.581	47.5	30.0	.923
world_religions	.826	42.2	27.1	.910
computer_security	.740	37.0	24.9	.908
college_physics	.436	31.6	30.1	.876
philosophy	.782	33.0	36.1	.868
abstract_algebra	.457	24.1	32.8	.822
us_foreign_policy	.864	20.4	36.9	.802
professional_medicine	.671	19.9	41.7	.778
high_school_math	.385	10.4	41.9	.614
Mean	—	31.1	32.6	.843

F FF Residual Variance Analysis

To investigate whether the elevated residual variance in free-form benchmarks (GSM8K 39.9%, MATH 43.9%) reflects unmodeled substantive sources or measurement properties of binary scoring, we regress item-level residual variance on item characteristics across all five benchmarks (Table 13).

The dominant predictor is the quadratic difficulty effect ($p < 10^{-5}$ on all benchmarks): items near $p = 0.5$ show maximal residual variance, while items near the floor or ceiling show minimal residual. This inverted-U pattern directly mirrors the Bernoulli variance ceiling $p(1-p)$. Item content features (number of reasoning steps, answer length) are not significant predictors for GSM8K ($p > 0.38$), confirming that the difficulty–residual relationship is statistical rather than content-driven.

FF benchmarks show substantially higher R^2 (mean 0.66) than MC benchmarks (mean 0.25), consistent with FF items spanning a wider difficulty range. The MC–FF contrast in residual magnitude is therefore a mathematical consequence of the difficulty distribution under binary scoring, not evidence of additional unmodeled variance sources in free-form evaluation.

Table 13: Item-level residual variance regression. Difficulty^2 = quadratic (inverted-U) term reflecting the Bernoulli $p(1-p)$ ceiling. GSM8K additionally tests n_steps and answer_length (both $p > 0.38$).

Benchmark	R^2 (full)	R^2 (diff. only)	Diff.^{2} p
Multiple-choice			
MMLU	0.377	0.347	<0.001
ARC	0.221	0.204	<0.001
HellaSwag	0.156	0.140	<0.001
Free-form			
GSM8K	0.468	0.449	<0.001
MATH	0.856	0.845	<0.001

G Model and Prompt Details

Models. See Table 3 (main text) for the eight models. Full HuggingFace identifiers: `meta-llama/Llama-3.1-8B-Instruct`, `google/gemma-2-9b-it`, `mistralai/Mistral-7B-Instruct-v0.3`, `Qwen/Qwen3-8B`, `01-ai/Yi-1.5-9B-Chat`, `allenai/OLMo-2-1124-7B-Instruct`, `internlm/internlm2_5-7b-chat`, `deepseek-ai/deepseek-llm-7b-chat`.

Prompt templates. Our six prompt templates follow a surface perturbation design inspired by BrittleBench (Romanou et al., 2026). For multiple-choice benchmarks, templates vary along three dimensions: option layout (lettered vs. numbered), instruction wording (direct vs. contextual), and answer prompt cue. For free-form benchmarks, templates vary chain-of-thought elicitation strategy. All six templates are semantically equivalent—they request the same task with different surface forms—to isolate formatting effects from content effects. ARC-Challenge and HellaSwag use the same six surface patterns as MMLU with domain-appropriate system instructions; MATH mirrors the GSM8K pattern with `\boxed{}` answer formatting.

MMLU templates (representative MC format):

1. `The following are multiple choice questions (with answers) about {subject}.\n\n{question}\nA. ... D. ...\nAnswer:`
2. `Answer the following {subject} questions by selecting the correct option.\nQuestion: {question}\n(A) ... (B) ... (C) ... (D) ...\nCorrect answer:`
3. `Below are {subject} multiple-choice problems. Pick the right letter.\nQ: {question}\n A) ... B) ... C) ... D) ...\nA:`
4. `Answer the following {subject} question by selecting A, B, C, or D.\n{question}\n- A. ... - B. ... - C. ... - D. ...\nAnswer:`
5. `Below is a {subject} exam question with four possible answers. Choose the correct one.\n{question}\nOption A: ... Option B: ... Option C: ... Option D: ...\nCorrect:`
6. `Question ({subject}):\n{question}\nChoice (A) ... (B) ... (C) ... (D) ...\nYour answer:`

GSM8K templates (representative FF format):

1. `Solve the following math problem step by step.\nQ: {question}\nA: Let’s think step by step.`
2. `Answer the following math word problem. Show your work.\nQuestion: {question}\nSolution:`
3. `Solve each problem below. Write your reasoning, then give the final answer after ####.\nProblem: {question}\nReasoning:`
4. `You are a math tutor. Solve the problem and show your steps.\n{question}\nSteps:`
5. `Work through this math question carefully.\nQ: {question}\nWork:`
6. `Math problem:\n{question}\nLet’s solve this step by step.`

H Use of AI Assistants

We used Claude (Anthropic) as an AI coding and writing assistant throughout this work. Specifically:

- **Code:** drafting experiment scripts, data-processing pipelines, and analysis utilities; debugging and refactoring.
- **Writing:** drafting and revising paper sections, polishing prose, and reformatting LaTeX.
- **Literature:** summarizing related papers to inform related-work section.